

Sleep Stage Classification Using Deep Learning

(BIO3609) Biomedical Signal Processing

Team 9

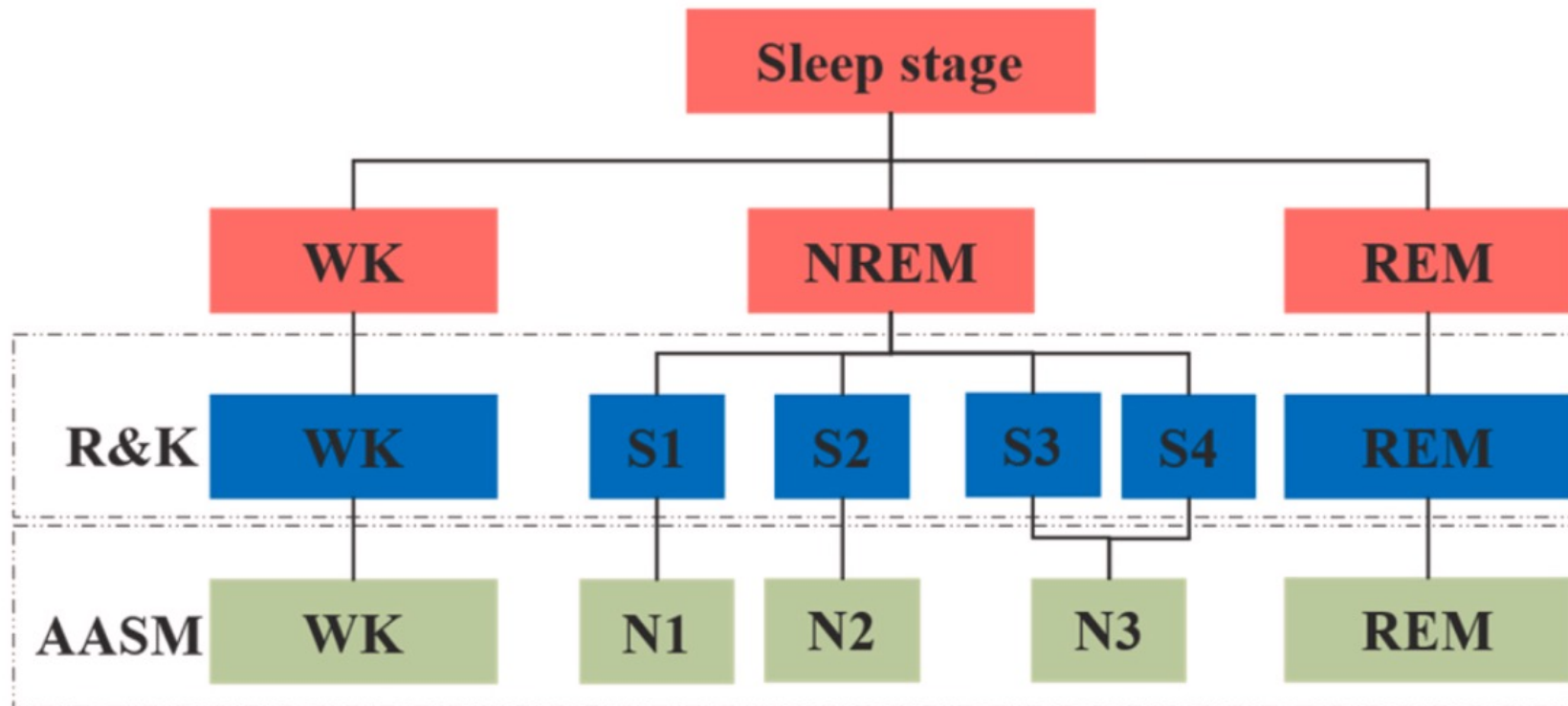
ليان عمرو – رقية مختار – هيا غراب

Problem Statement

- Sleep is divided into multiple stages.
- Sleep staging is important for diagnosing sleep disorders (e.g. sleep apnea).
- Manual scoring by experts is time-consuming.
- Deep learning can automatically classify sleep stages thus saving us time and effort.

Sleep Stages

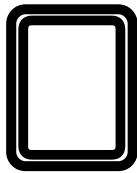
EEG, EOG and EMG signals are used for classification.



Dataset

MESA Sleep Study Dataset:

- Part of the Multi-Ethnic Study of Atherosclerosis.
- Around 2,237 adults ages 45-84 years old.
- Collected from 2010-2013 from multi-ethnic populations.
- Uses overnight polysomnography recording.
- Data includes EEG, EOG, EMG, ECG, respiratory signals and oxygen saturations.
- Divided into signals (.edf) and annotations (.xml).

Research Paper 

“A Multi-task Deep Learning Algorithm for Sleep Stage Scoring and Sleep Arousal Detection”

Zhang, X., Wang, Z., Yang, Y., Li, X., & Zhang, J. (2023)

Idea and dataset

Develop a multi-task deep learning model that predicts:

- Sleep stages
- Sleep arousal

Using only ECG signals.

MESA Sleep Dataset split:

64% training set, 16% validation set, 20% testing set.

ECG sampling rate: 256 Hz.

Classification accuracy: 79%

Model Architecture

Model type:

Hybrid CNN+LSTM deep learning architecture.

Arousal is binary labeled

1 = arousal

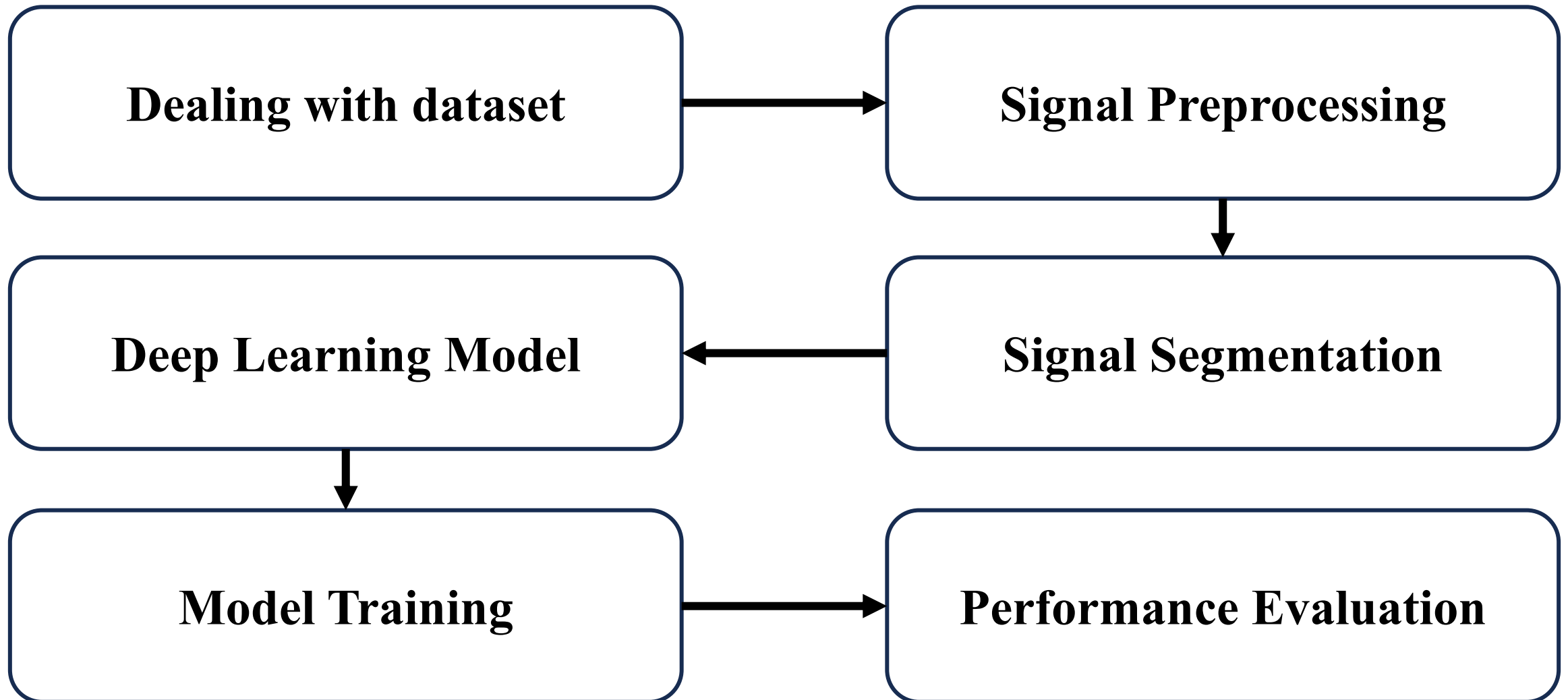
0 = no arousal

Sleep stage classes:

- Wake
- Light sleep (N1+N2)
- Deep sleep (N3)
- REM

Project Pipeline 🙌

Pipeline

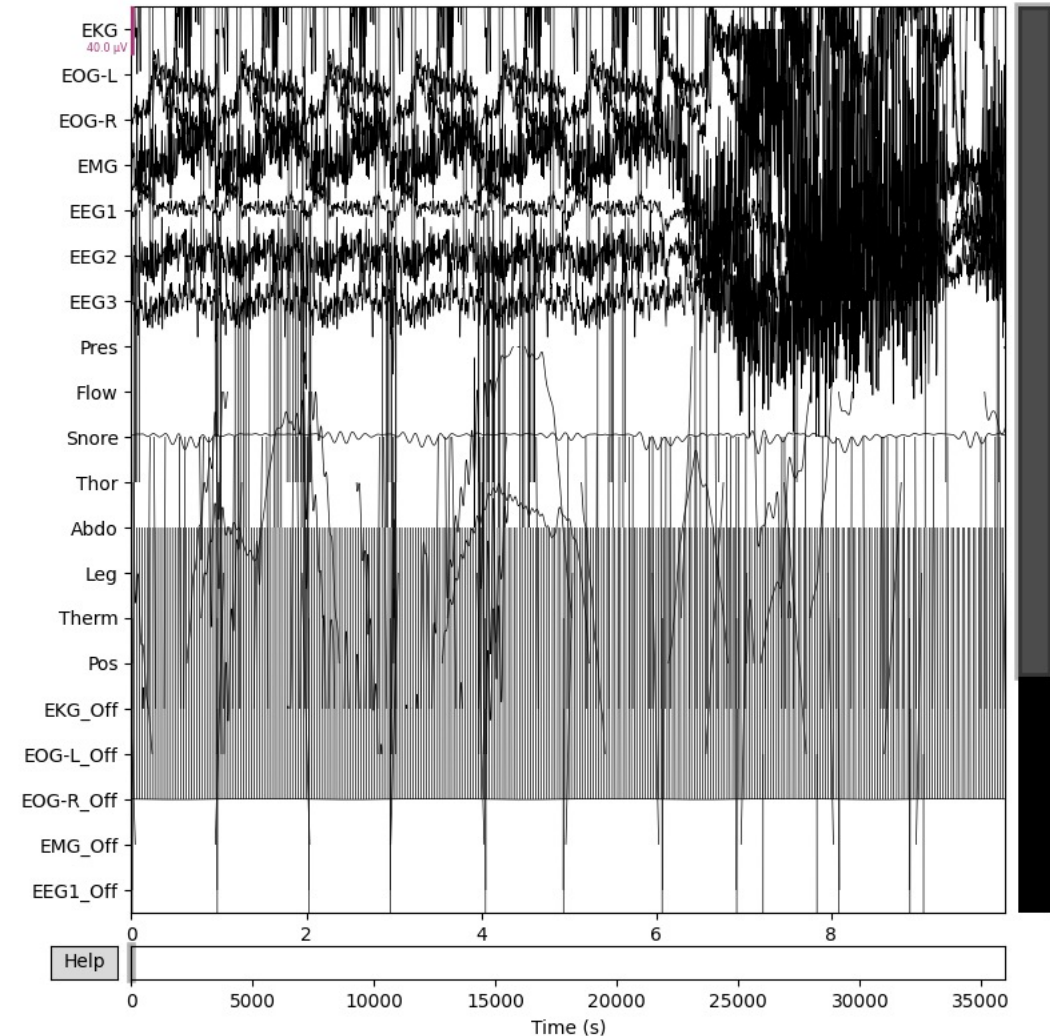


Dealing with dataset

- Load sleep recordings.
- Select the signals needed.

Printed a sample of the signal:

- Each horizontal line = one signal channel.
- Time axis shows entire night ~10 hours.
- Sample frequency: 256 Hz.



Signal Preprocessing

To improve signal quality, we will use these steps:

- Filtering (to remove physiological signals ‘noise’)
- Artifact removal
- Normalization

Signal Segmentation

Sleep recordings are divided into 30 second segments called epochs.

Full night recording → 30 second epochs → each epoch gets a sleep stage label.

Labels: Wake, N1, N2, N3, REM.

Deep Learning Model

The model learns patterns from signal.

Suggested architecture:

- CNN
- CNN + LSTM
- CNN + Transformer

Model Training

- Train the model using labeled epochs.
- Split dataset into:
 - training set
 - validation set
 - test set

Performance Evaluation

The model's performance is measured using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Thank you ☺